

Generalization of a Sequence Limit Involving Riemann Sums and Perturbations

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August 11, 2026

Here is a rigorous generalization of a class of limit problems where the terms of a sequence can be separated into a dominant Riemann sum and a vanishing error term. We can generalize this technique to any strictly positive, continuous function combined with an asymptotically vanishing perturbation.

General Theorem

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $f(x) > 0$ for all $x \in [0, 1]$. Let $g_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of functions such that $\lim_{n \rightarrow \infty} \sup_{x \in [0, 1]} |g_n(x)| = 0$.

Define the sequence $(u_n)_{n \in \mathbb{N}^*}$ by:

$$u_n = \left(\prod_{k=1}^n \left(f\left(\frac{k}{n}\right) + g_n\left(\frac{k}{n}\right) \right) \right)^{\frac{1}{n}}$$

Then, the limit of the sequence is:

$$\lim_{n \rightarrow \infty} u_n = \exp \left(\int_0^1 \ln(f(x)) dx \right)$$

Rigorous Proof

1. Taking the Natural Logarithm

The standard approach for a sequence involving an n -th root of a product is to examine its natural logarithm:

$$\ln(u_n) = \frac{1}{n} \sum_{k=1}^n \ln \left(f \left(\frac{k}{n} \right) + g_n \left(\frac{k}{n} \right) \right)$$

Because $f(x) > 0$ for all $x \in [0, 1]$, we can factor out $f \left(\frac{k}{n} \right)$:

$$\ln(u_n) = \frac{1}{n} \sum_{k=1}^n \ln \left(f \left(\frac{k}{n} \right) \left[1 + \frac{g_n \left(\frac{k}{n} \right)}{f \left(\frac{k}{n} \right)} \right] \right)$$

Using the logarithmic property $\ln(ab) = \ln(a) + \ln(b)$, we split this into two sums:

$$\ln(u_n) = \frac{1}{n} \sum_{k=1}^n \ln \left(f \left(\frac{k}{n} \right) \right) + \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{g_n \left(\frac{k}{n} \right)}{f \left(\frac{k}{n} \right)} \right)$$

2. Bounding the Error Term

Because f is continuous and strictly positive on the closed, bounded interval $[0, 1]$, by the Extreme Value Theorem, f attains a minimum value $m > 0$. Therefore, for all $x \in [0, 1]$, we have:

$$\left| \frac{g_n(x)}{f(x)} \right| \leq \frac{|g_n(x)|}{m} \leq \frac{1}{m} \sup_{x \in [0, 1]} |g_n(x)|$$

Since $\sup_{x \in [0, 1]} |g_n(x)| \rightarrow 0$ as $n \rightarrow \infty$, there exists some N such that for all $n > N$, $\left| \frac{g_n(x)}{f(x)} \right| \leq \frac{1}{2}$. For $|y| \leq \frac{1}{2}$, we can use the Taylor series bound $|\ln(1 + y)| \leq 2|y|$. Thus, we establish an upper bound for the error term:

$$\left| \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{g_n \left(\frac{k}{n} \right)}{f \left(\frac{k}{n} \right)} \right) \right| \leq \frac{1}{n} \sum_{k=1}^n 2 \left| \frac{g_n \left(\frac{k}{n} \right)}{f \left(\frac{k}{n} \right)} \right| \leq \frac{2}{m} \sup_{x \in [0, 1]} |g_n(x)|$$

By the Squeeze Theorem, since the right side converges to 0 as $n \rightarrow \infty$, the error term vanishes.

3. Evaluating the Riemann Sum

With the error term vanishing, the limit is entirely determined by the first sum:

$$\lim_{n \rightarrow \infty} \ln(u_n) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \left(f \left(\frac{k}{n} \right) \right)$$

Because f is continuous, $\ln \circ f$ is also continuous on $[0, 1]$. Therefore, this standard Riemann sum converges to the definite integral:

$$\lim_{n \rightarrow \infty} \ln(u_n) = \int_0^1 \ln(f(x)) dx$$

Exponentiating the result to find the limit of the original sequence u_n , we conclude:

$$\lim_{n \rightarrow \infty} u_n = \exp \left(\int_0^1 \ln(f(x)) dx \right)$$

Application to a Specific Problem

Consider the sequence where the general term in the product is $1 + \frac{k}{n} + \frac{k}{n^2}$. We define our functions as:

- $f(x) = 1 + x$
- $g_n(x) = \frac{x}{n}$

Here, the term $\frac{k}{n^2}$ corresponds exactly to $g_n \left(\frac{k}{n} \right) = \frac{1}{n} \left(\frac{k}{n} \right)$. We can easily verify the conditions of the generalization:

1. $f(x) = 1 + x > 0$ for all $x \in [0, 1]$.
2. $\sup_{x \in [0, 1]} |g_n(x)| = \sup_{x \in [0, 1]} \frac{x}{n} = \frac{1}{n}$, which approaches 0 as $n \rightarrow \infty$.

Applying our generalized theorem, the limit is given directly by:

$$\exp \left(\int_0^1 \ln(1 + x) dx \right)$$

Evaluating the integral using integration by parts yields $2 \ln(2) - 1$, leading to the final result of $\frac{4}{e}$.